Problem Set 3

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Due: November 12, 2026

Instructions

- Please show your work! You may lose points by simply writing in the answer. If the problem requires you to execute commands in `R`, please include the code you used to get your answers. Please also include the `.R` file that contains your code. If you are not sure if work needs to be shown for a particular problem, please ask.
- Your homework should be submitted electronically on GitHub.
- This problem set is due before 23:59 on Thursday November 12, 2026. No late assignments will be accepted.

In this problem set, you will run several regressions and create an add variable plot (see the lecture slides) in `R` using the `incumbents_subset.csv` dataset. Include all of your code.

Question 1

We are interested in knowing how the difference in campaign spending between incumbent and challenger affects the incumbent's vote share.

1. Run a regression where the outcome variable is `voteshare` and the explanatory variable is `difflog`.

   I regress *voteshare* on *difflog* using the following commands in R:

```
reg_voteshare_difflog <- lm(inc.sub$voteshare ~ inc.sub$difflog, data=inc.sub)
reg_voteshare_difflog
summary(reg_voteshare_difflog)
```

   The "summary" command produces the following output:

```
Residuals:
Min       1Q   Median       3Q      Max
-0.26832 -0.05345 -0.00377  0.04780  0.32749

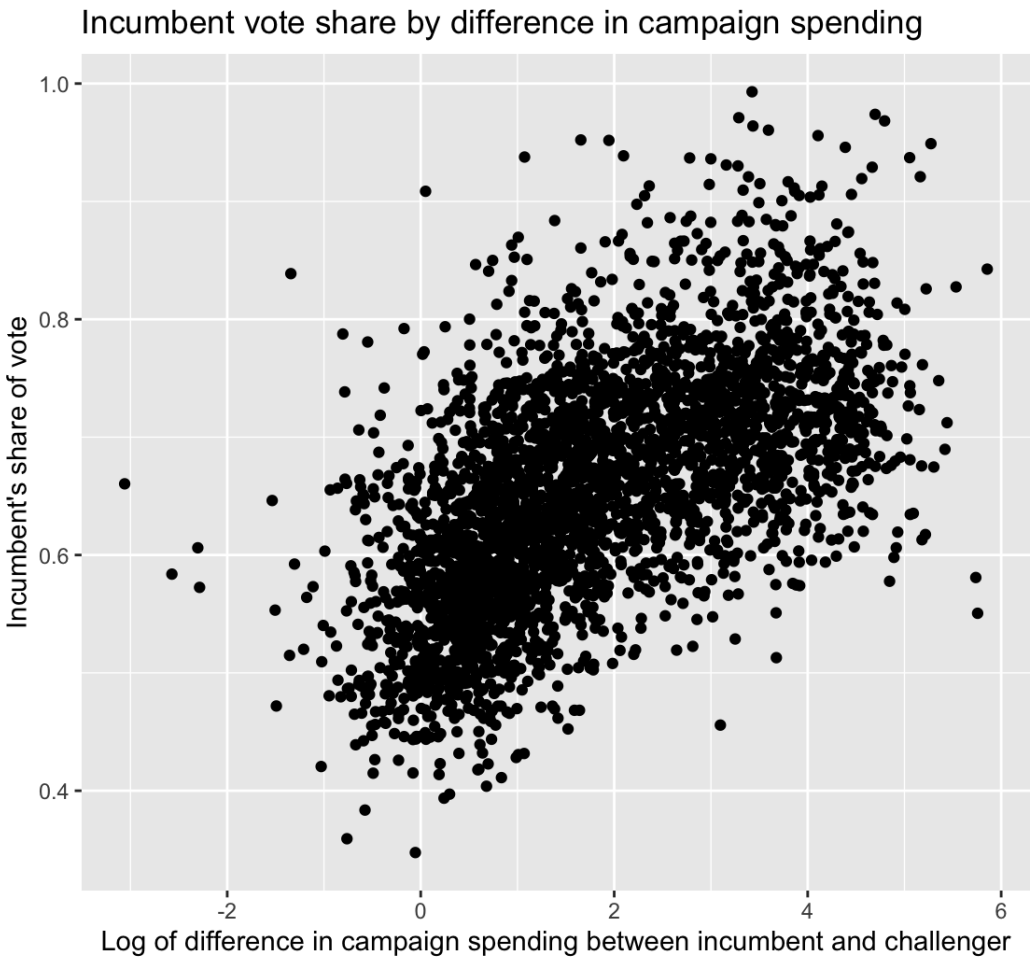
Coefficients:
Estimate Std. Error t value
(Intercept)     0.579031   0.002251  257.19
inc.sub$difflog 0.041666   0.000968   43.04
Pr(>|t|)
(Intercept)       <2e-16 ***
inc.sub$difflog   <2e-16 ***
---
Signif. codes:
0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.07867 on 3191 degrees of freedom
Multiple R-squared:  0.3673, Adjusted R-squared:  0.3671
F-statistic:  1853 on 1 and 3191 DF,  p-value: < 2.2e-16
```

2. Make a scatterplot of the two variables and add the regression line.

   First, I create the scatterplot using the following code:

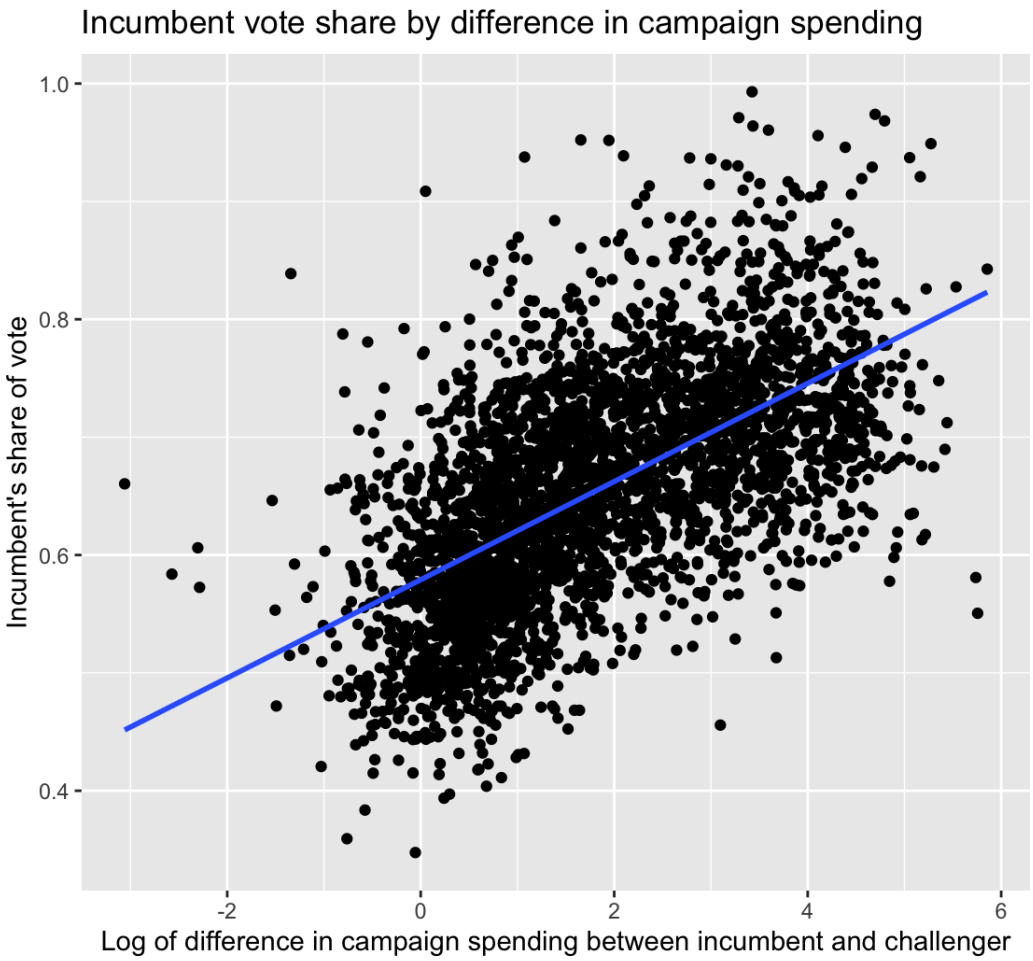
```r
library(ggplot2)
ggplot(inc.sub, aes(x=difflog, y=voteshare)) +
  geom_point() +
  labs(title = "Incumbent vote share by difference in campaign spending",
       x = "Log of difference in campaign spending between incumbent and challenger",
       y = "Incumbent's share of vote")
```



The scatterplot shows a moderate, positive association between the log of the difference in campaign spending between incumbent and challenger and the incumbent's share of the vote. There do not appear to be any outliers in the data.

I add the regression line using this code:

```
ggplot(inc.sub, aes(x=difflog, y=voteshare)) +
  geom_point() +
  labs(title = "Incumbent vote share by difference in campaign spending",
       x = "Log of difference in campaign spending between incumbent and
    challenger",
       y = "Incumbent's share of vote") +
  geom_smooth(
    aes(x=difflog, y=voteshare),
    method = "lm",
    formula = y~x,
    se=FALSE)
```



From the summary table above, the regression line equation is:

$$\hat{voteshare} = \mathbf{0.579031} + \mathbf{0.041666}(\boldsymbol{difflog}) + \epsilon$$

An incumbent for whom the log of the difference in campaign spending between them and the challenger is zero would be expected to, on average, receive a 0.579031 share of the vote. For every one-unit increase in the log of the difference in spending between incumbent and challenger, the incumbent would expect to, on average, receive an additional 0.041666 share of the vote.

3. Save the residuals of the model in a separate object.

   I save the residuals of the model in a separate object using this command:

```
residuals_voteshare_difflog <- resid(reg_voteshare_difflog)
```

4. Write the prediction equation.

   I again run the "summary" command to get the regression coefficient values:

```r
summary(reg_voteshare_difflog)
```

```
Residuals:
Min       1Q   Median       3Q      Max
-0.26832 -0.05345 -0.00377  0.04780  0.32749

Coefficients:
Estimate Std. Error t value
(Intercept)     0.579031   0.002251  257.19
inc.sub$difflog 0.041666   0.000968   43.04
Pr(>|t|)
(Intercept)       <2e-16 ***
inc.sub$difflog   <2e-16 ***
---
Signif. codes:
0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.07867 on 3191 degrees of freedom
Multiple R-squared:  0.3673, Adjusted R-squared:  0.3671
F-statistic:  1853 on 1 and 3191 DF,  p-value: < 2.2e-16
```

From the output above:

$\hat{\beta}_0$ (intercept) $= 0.579031$

$\hat{\beta}_1$ (slope) $= 0.041666$

Therefore:

$$\hat{voteshare} = \mathbf{0.579031 + 0.041666(\textit{difflog})} + \epsilon$$

An incumbent for whom the log of the difference in campaign spending between them and the challenger is zero would be expected to, on average, receive a 0.579031 share of the vote. For every one-unit increase in the log of the difference in spending between incumbent and challenger, the incumbent would expect to, on average, receive an additional 0.041666 share of the vote.

Question 2

We are interested in knowing how the difference between incumbent and challenger's spending and the vote share of the presidential candidate of the incumbent's party are related.

1. Run a regression where the outcome variable is `presvote` and the explanatory variable is `difflog`.

   I regress *presvote* on *difflog* using the below code:

```
reg_presvote_difflog <- lm(inc.sub$presvote ~ inc.sub$difflog, data=inc.sub)
reg_presvote_difflog
summary(reg_presvote_difflog)
```

   The "summary" command produces the following output:

```
Residuals:
Min       1Q   Median       3Q      Max
-0.32196 -0.07407 -0.00102  0.07151  0.42743

Coefficients:
Estimate Std. Error t value Pr(>|t|)
(Intercept)     0.507583   0.003161  160.60   <2e-16 ***
inc.sub$difflog 0.023837   0.001359   17.54   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

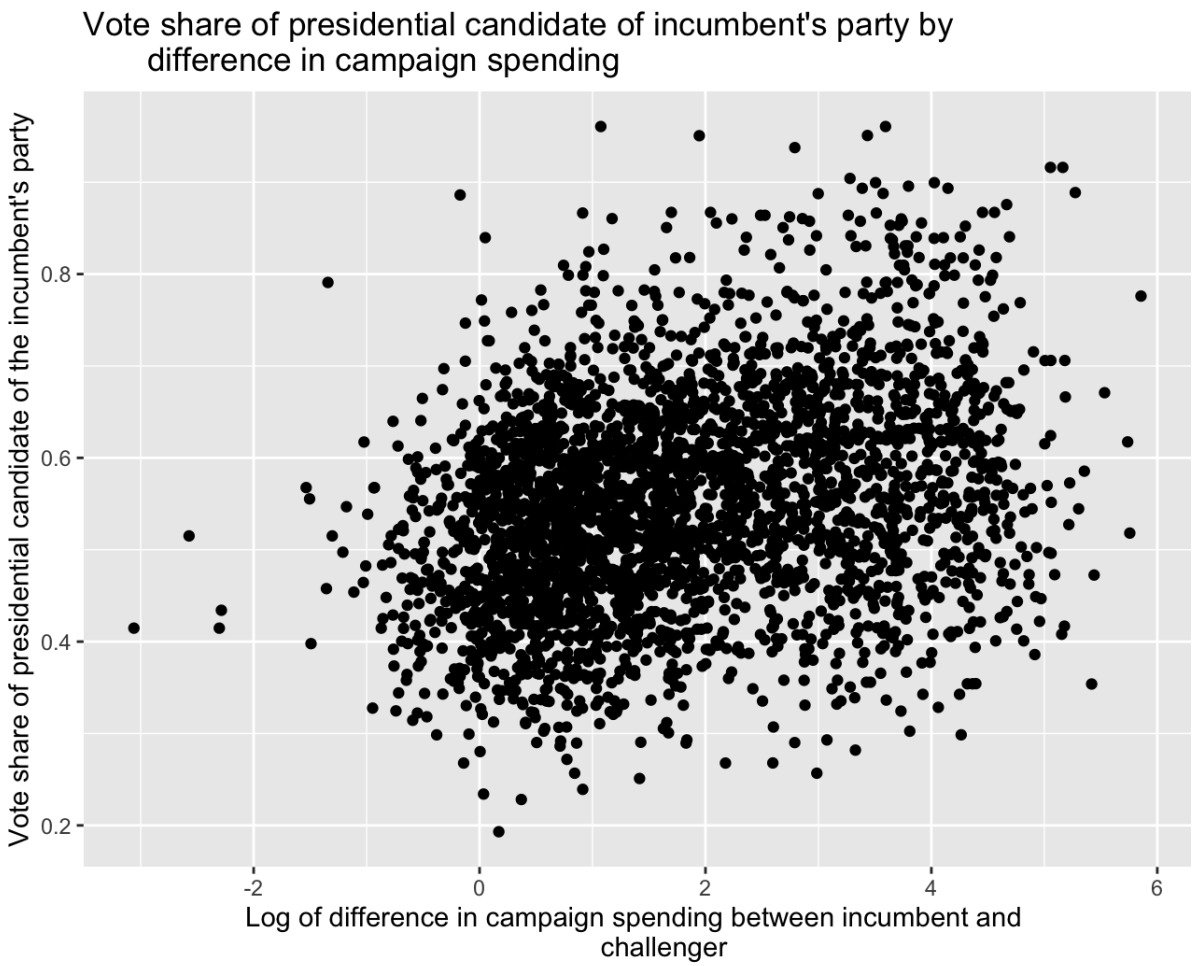
Residual standard error: 0.1104 on 3191 degrees of freedom
Multiple R-squared:  0.08795, Adjusted R-squared:  0.08767
F-statistic: 307.7 on 1 and 3191 DF,  p-value: < 2.2e-16
```

2. Make a scatterplot of the two variables and add the regression line.

   First, I create the scatterplot using the following code:

```
library(ggplot2)
ggplot(inc.sub, aes(x=difflog, y=presvote)) +
  geom_point() +
  labs(title = "Vote share of presidential candidate of incumbent's party by
      difference in campaign spending",
      x = "Log of difference in campaign spending between incumbent and
      challenger",
```

```
8         y = "Vote share of presidential candidate of the incumbent's party
    ")
```

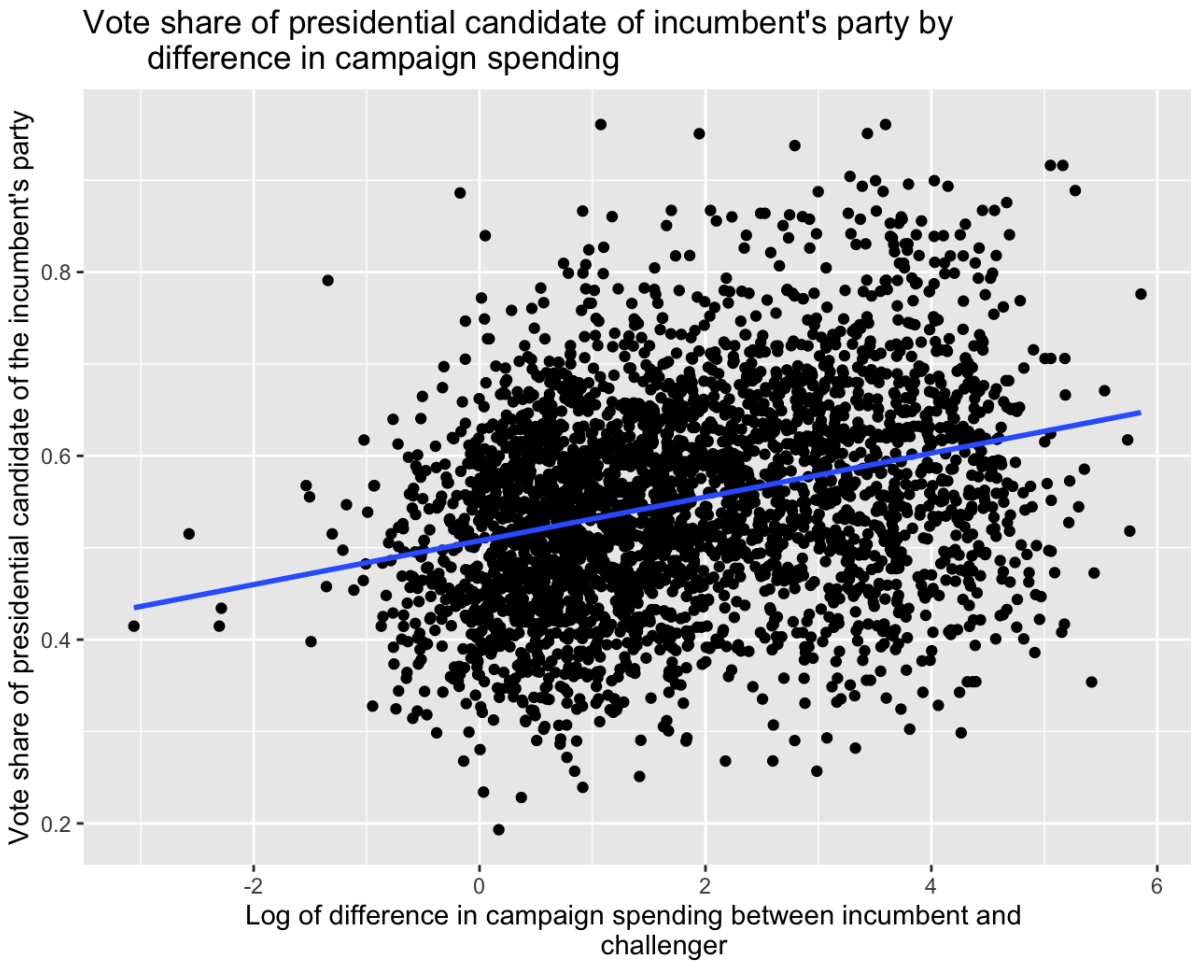


The scatterplot shows a weak, positive association between the log of the difference in campaign spending between challenger and incumbent and the vote share received by the presidential candidate of the incumbent's party.

I add the regression line using this code:

```
ggplot(inc.sub, aes(x=difflog, y=presvote)) +
  geom_point() +
  labs(title = "Vote share of presidential candidate of incumbent's party
     by
        difference in campaign spending",
        x = "Log of difference in campaign spending between incumbent and
        challenger",
        y = "Vote share of presidential candidate of the incumbent's party
    ") +
  geom_smooth(
    aes(x=difflog, y=presvote),
```

```
    method = "lm",
    formula = y~x,
    se=FALSE)
```



From the summary table above, the regression line equation is:

$$\hat{presvote} = \mathbf{0.507583} + \mathbf{0.023837}(\boldsymbol{difflog}) + \epsilon$$

On average, a presidential candidate of the incumbent's party for whom the log of the difference in campaign spending between the incumbent and the challenger is zero would be expected to, on average, receive a 0.507583 share of the vote. For every one-unit increase in the log of the difference in spending between incumbent and challenger, the presidential candidate of the incumbent's party would expect to, on average, receive an additional 0.023837 share of the vote.

3. Save the residuals of the model in a separate object.

I save the residuals of the model in a separate object using this command:

```
residuals_presvote_difflog <- resid(reg_presvote_difflog)
```

4. Write the prediction equation.

   I again run the "summary" command to get the regression coefficient values:

```
summary(reg_presvote_difflog)
```

   The "summary" command produces the following output:

```
Residuals:
Min       1Q   Median       3Q      Max
-0.32196 -0.07407 -0.00102  0.07151  0.42743

Coefficients:
Estimate Std. Error t value Pr(>|t|)
(Intercept)     0.507583   0.003161  160.60   <2e-16 ***
inc.sub$difflog 0.023837   0.001359   17.54   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1104 on 3191 degrees of freedom
Multiple R-squared:  0.08795, Adjusted R-squared:  0.08767
F-statistic: 307.7 on 1 and 3191 DF,  p-value: < 2.2e-16
```

   From the output above:

   $\hat{\beta}_0$ (intercept) = 0.507583

   $\hat{\beta}_1$ (slope) = 0.023837

   Therefore:

   $\hat{presvote} = \mathbf{0.507583 + 0.023837(\textit{difflog})} + \epsilon$

   On average, a presidential candidate of the incumbent's party for whom the log of the difference in campaign spending between the incumbent and the challenger is zero would be expected to, on average, receive a 0.507583 share of the vote. For every one-unit increase in the log of the difference in spending between incumbent and challenger, the presidential candidate of the incumbent's party would expect to, on average, receive an additional 0.023837 share of the vote.

Question 3

We are interested in knowing how the vote share of the presidential candidate of the incumbent's party is associated with the incumbent's electoral success.

1. Run a regression where the outcome variable is `voteshare` and the explanatory variable is `presvote`.

   I regress *voteshare* on *presvote* using the following code:

```
reg_voteshare_presvote <- lm(inc.sub$voteshare ~ inc.sub$presvote, data=
    inc.sub)
reg_voteshare_presvote
summary(reg_voteshare_presvote)
```

   The "summary" command produces the following output:

```
Residuals:
Min       1Q   Median       3Q      Max
-0.27330 -0.05888  0.00394  0.06148  0.41365

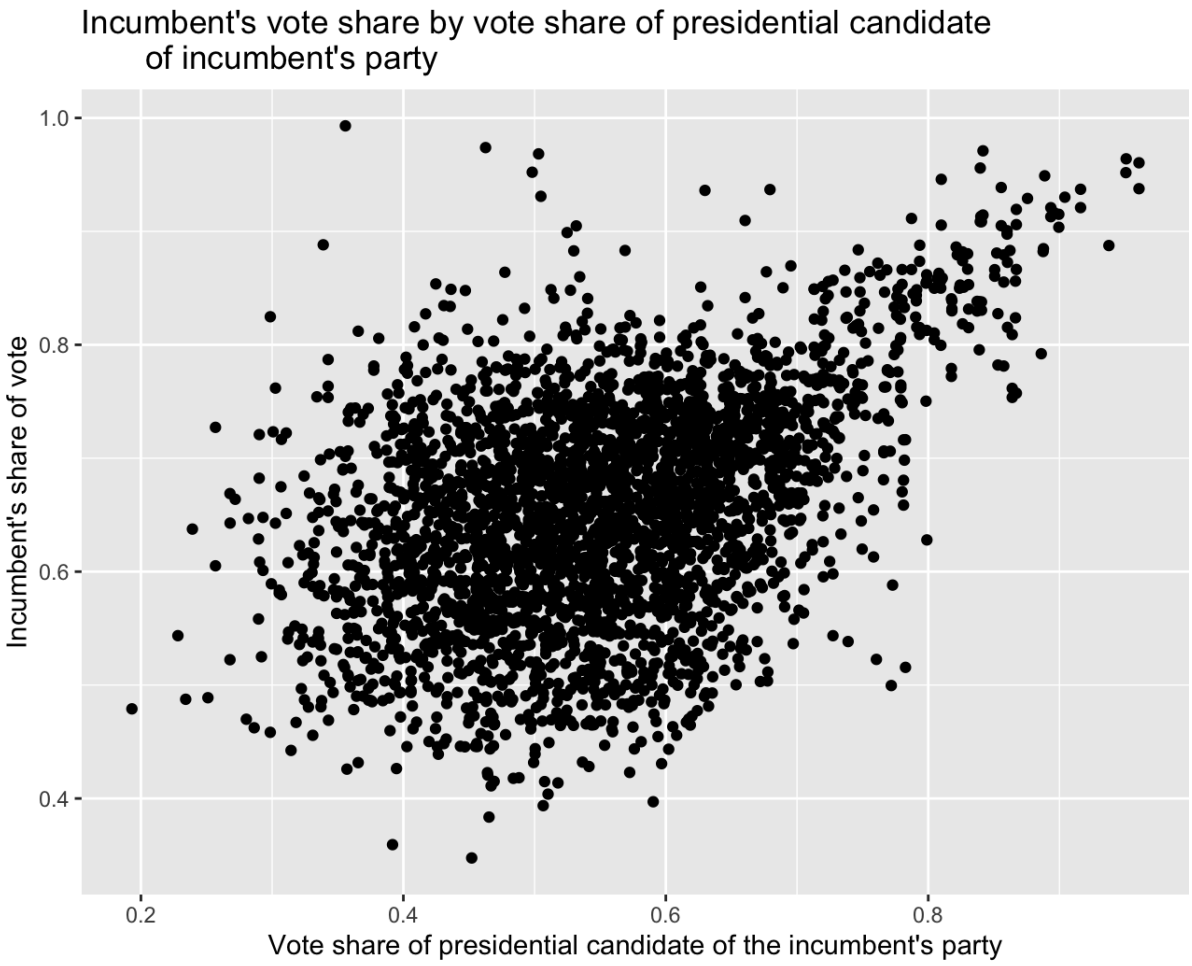
Coefficients:
Estimate Std. Error t value Pr(>|t|)
(Intercept)      0.441330   0.007599   58.08   <2e-16 ***
inc.sub$presvote 0.388018   0.013493   28.76   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.08815 on 3191 degrees of freedom
Multiple R-squared:  0.2058, Adjusted R-squared:  0.2056
F-statistic:   827 on 1 and 3191 DF,  p-value: < 2.2e-16
```

2. Make a scatterplot of the two variables and add the regression line.

   First, I generate the scatterplot using the following code:

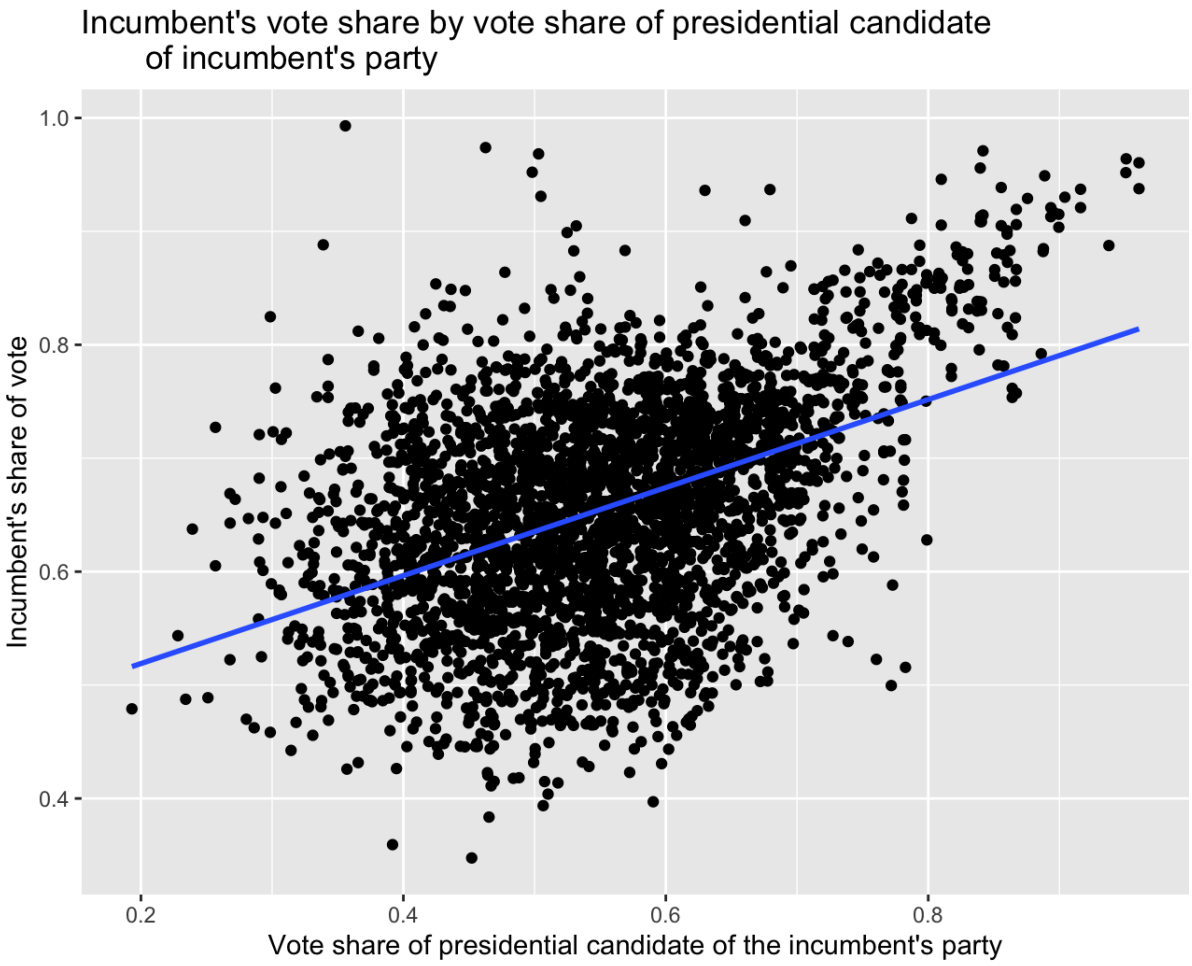
```
library(ggplot2)
ggplot(inc.sub, aes(x=presvote, y=voteshare)) +
  geom_point() +
  labs(title = "Incumbent's vote share by vote share of presidential
    candidate
       of incumbent's party",
       x = "Vote share of presidential candidate of the incumbent's party
    ",
       y = "Incumbent's share of vote")
```



The scatterplot shows a moderate, positive association between the vote share received by the presidential candidate of the incumbent's party and the vote share received by the incumbent.

I add the regression line using the following code:

```
ggplot(inc.sub, aes(x=presvote, y=voteshare)) +
  geom_point() +
  labs(title = "Incumbent's vote share by vote share of presidential
   candidate
       of incumbent's party",
       x = "Vote share of presidential candidate of the incumbent's party
   ",
       y = "Incumbent's share of vote") +
  geom_smooth(
    aes(x=presvote, y=voteshare),
    method = "lm",
    formula = y~x,
    se=FALSE)
```



From the output table above, the regression line equation is:

$$\hat{voteshare} = \mathbf{0.441330} + \mathbf{0.388018}(\boldsymbol{presvote}) + \epsilon$$

On average, an incumbent for whom the presidential candidate of their own party received no share of the votes can expect to themselves receive a 0.441330 share of the vote. For a one-unit increase in the presidential candidate's vote share (i.e., the candidate moves from 0 votes to a 1.0, or 100 percent share of the votes), the incumbent would expect to, on average, receive an additional 0.388018 share of the votes.

3. Write the prediction equation.

   I again run the "summary" command to get the regression coefficient values:

```
summary(reg_voteshare_presvote)
```

   The "summary" command produces the following output:

```
Residuals:
Min       1Q   Median       3Q      Max
```

```
-0.27330 -0.05888  0.00394  0.06148  0.41365

Coefficients:
Estimate Std. Error t value Pr(>|t|)
(Intercept)      0.441330   0.007599   58.08   <2e-16 ***
inc.sub$presvote 0.388018   0.013493   28.76   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.08815 on 3191 degrees of freedom
Multiple R-squared:  0.2058, Adjusted R-squared:  0.2056
F-statistic:   827 on 1 and 3191 DF,  p-value: < 2.2e-16
```

From the output above:

$\hat{\beta}_0$ (intercept) = 0.441330

$\hat{\beta}_1$ (slope) = 0.388018

Therefore:

$$\hat{voteshare} = \mathbf{0.441330 + 0.388018(\boldsymbol{presvote})} + \epsilon$$

On average, an incumbent for whom the presidential candidate of their own party received no share of the votes can expect to themselves receive a 0.441330 share of the vote. For a one-unit increase in the presidential candidate's vote share (i.e., the candidate moves from 0 votes to a 1.0, or 100 percent share of the votes), the incumbent would expect to, on average, receive an additional 0.388018 share of the votes.

Question 4

The residuals from part (a) tell us how much of the variation in `voteshare` is *not* explained by the difference in spending between incumbent and challenger. The residuals in part (b) tell us how much of the variation in `presvote` is *not* explained by the difference in spending between incumbent and challenger in the district.

1. Run a regression where the outcome variable is the residuals from Question 1 and the explanatory variable is the residuals from Question 2.

   I regress (the residuals from regressing *voteshare* on *difflog*) on (the residuals from regressing *presvote* on *difflog*) using the following code:

```
reg_residuals <- lm(residuals_voteshare_difflog ~ residuals_presvote_
    difflog)
reg_residuals
summary(reg_residuals)
```

   The "summary" command produces the following output:

```
Residuals:
Min       1Q   Median       3Q      Max
-0.25928 -0.04737 -0.00121  0.04618  0.33126

Coefficients:
Estimate Std. Error t value Pr(>|t|)
(Intercept)                -1.942e-18  1.299e-03    0.00        1
residuals_presvote_difflog  2.569e-01  1.176e-02   21.84   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

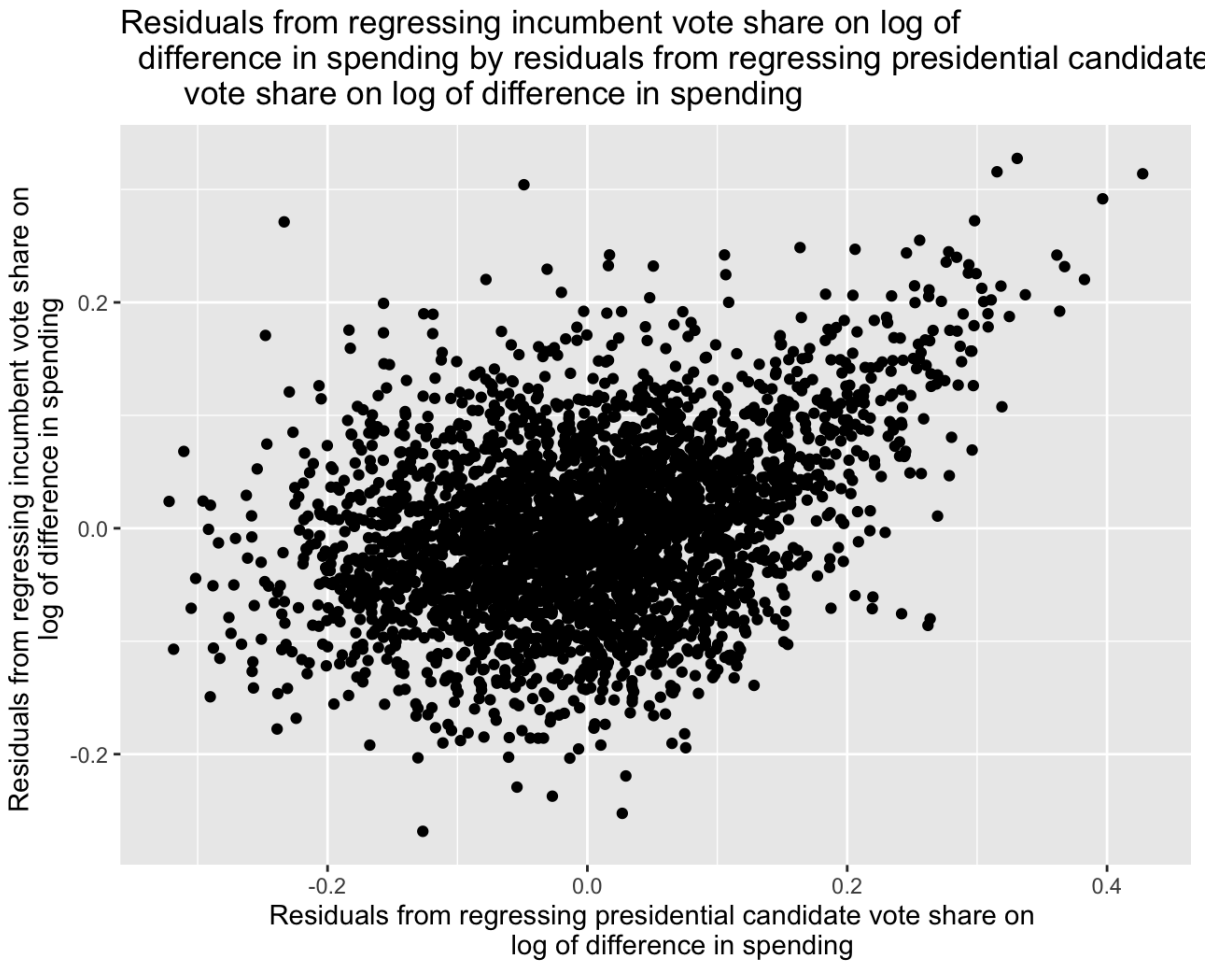
Residual standard error: 0.07338 on 3191 degrees of freedom
Multiple R-squared:   0.13, Adjusted R-squared:  0.1298
F-statistic:   477 on 1 and 3191 DF,  p-value: < 2.2e-16
```

2. Make a scatterplot of the two residuals and add the regression line.

   First, I generate a scatterplot using the following code:

```
library(ggplot2)
ggplot(inc.sub, aes(x=residuals_presvote_difflog, y=residuals_voteshare_
    difflog)) +
  geom_point() +
  labs(title = "Residuals from regressing incumbent vote share on log of
```

```
  difference in spending by residuals from regressing presidential
    candidate
      vote share on log of difference in spending",
      x = "Residuals from regressing presidential candidate vote share
    on
      log of difference in spending",
      y = "Residuals from regressing incumbent vote share on
      log of difference in spending")bboxmarginendforaddto also
```

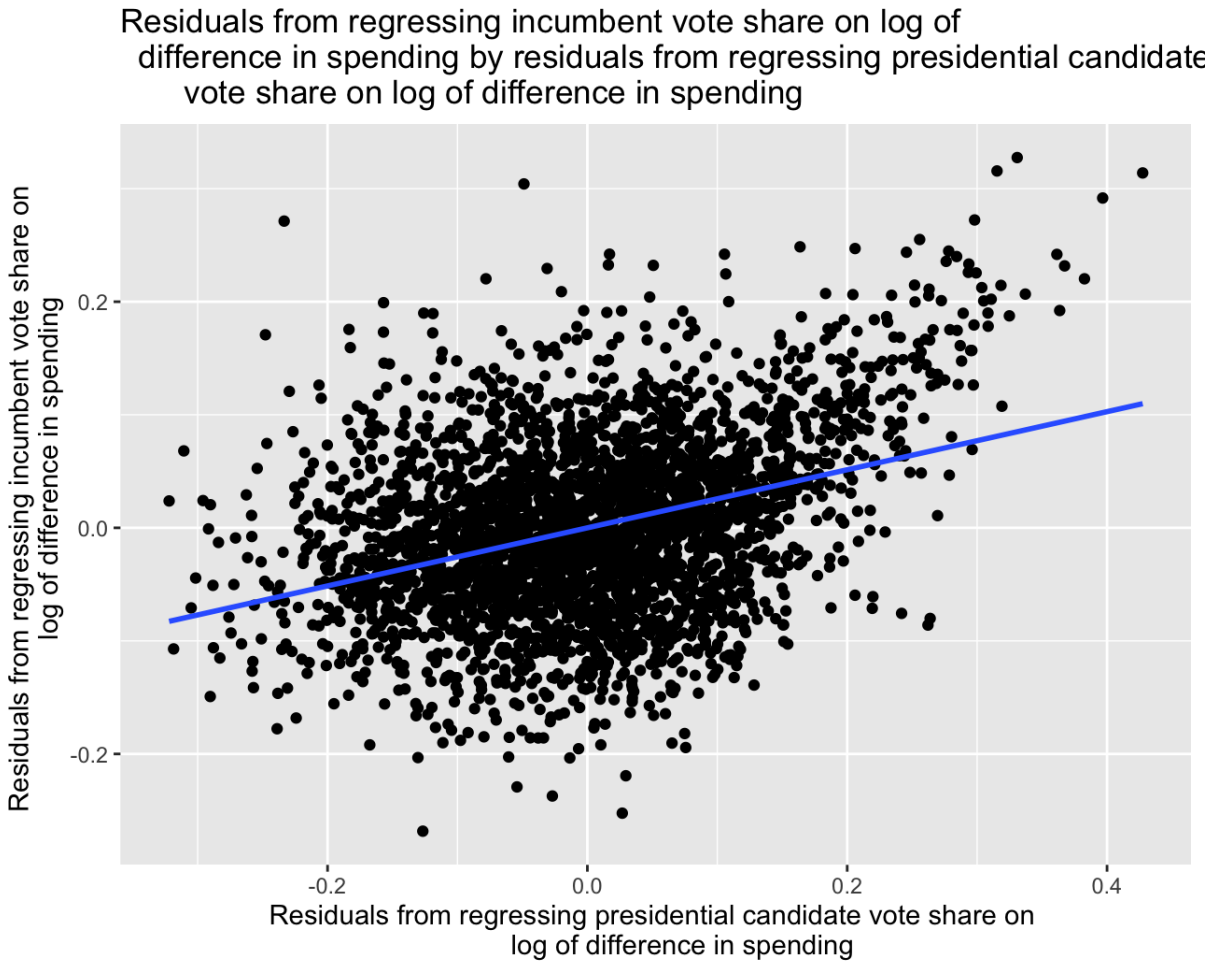


The scatterplot shows a moderate, positive association between the residuals from regressing the presidential candidate vote share on the log of the difference in spending and the residuals from regressing incumbent vote share on the log of the difference in spending.

I add the regression line using the following code:

```
ggplot(inc.sub, aes(x=residuals_presvote_difflog, y=residuals_voteshare_
    difflog)) +
  geom_point() +
  labs(title = "Residuals from regressing incumbent vote share on log of
```

```
  difference in spending by residuals from regressing presidential
   candidate
      vote share on log of difference in spending",
      x = "Residuals from regressing presidential candidate vote share
   on
      log of difference in spending",
      y = "Residuals from regressing incumbent vote share on
      log of difference in spending") +
 geom_smooth(
   aes(x=residuals_presvote_difflog, y=residuals_voteshare_difflog),
   method = "lm",
   formula = y~x,
   se=FALSE)
```



From the summary table above, the regression line equation is:

$\widehat{residual\,from\,regressing\,voteshare\,on\,difflog} = \textbf{(-1.942e-18)} + \textbf{(2.569e-01)}(\textbf{\textit{residual from regressing presvote on difflog}}) + \epsilon$

NOTE: To simplify the below interpretation, I will use *voteshare* as shorthand for "incumbent's share of the vote," *presvote* as shorthand for "vote share of presidential candidate of the incumbent's party," and *difflog* as shorthand for "log of difference in campaign spending between challenger and incumbent."

INTERPRETATION: On average, we could expect the residual from regressing *voteshare* on *difflog* for a data point where the residual from regressing *presvote* on *difflog* is zero to be -1.942e-18, i.e., close to zero. For a one-unit increase in the residual from regressing *presvote* on *difflog*, we can expect the residual from regressing *voteshare* on *difflog* for that data point to increase, on average, by 2.569e-01.

3. Write the prediction equation.

   I run the "summary" command again to get the regression coefficient values:

```
summary(reg_residuals)
```

   The "summary" command produces the following output:

```
Residuals:
Min       1Q   Median       3Q      Max
-0.25928 -0.04737 -0.00121  0.04618  0.33126

Coefficients:
Estimate Std. Error t value Pr(>|t|)
(Intercept)                -1.942e-18  1.299e-03    0.00        1
residuals_presvote_difflog  2.569e-01  1.176e-02   21.84   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.07338 on 3191 degrees of freedom
Multiple R-squared:   0.13, Adjusted R-squared:  0.1298
F-statistic:   477 on 1 and 3191 DF,  p-value: < 2.2e-16
```

From the output above:

$\hat{\beta}_0$ (intercept) = -1.942e-18

$\hat{\beta}_1$ (slope) = 2.569e-01

Therefore:

$\hat{residual from regressing voteshare on difflog}$ = **(-1.942e-18) + (2.569e-01)(*residual from regressing presvote on difflog*)** + ϵ

NOTE: To simplify the below interpretation, I will use *voteshare* as shorthand for "incumbent's share of the vote," *presvote* as shorthand for "vote share of presidential candidate of the incumbent's party," and *difflog* as shorthand for "log of difference in campaign spending between challenger and incumbent."

INTERPRETATION: On average, we could expect the residual from regressing *voteshare* on *difflog* for a data point where the residual from regressing *presvote* on *difflog* is zero to be -1.942e-18, i.e., close to zero. For a one-unit increase in the residual from regressing *presvote* on *difflog*, we can expect the residual from regressing *voteshare* on *difflog* for that data point to increase, on average, by 2.569e-01.

Question 5

What if the incumbent's vote share is affected by both the president's popularity and the difference in spending between incumbent and challenger?

1. Run a regression where the outcome variable is the incumbent's `voteshare` and the explanatory variables are `difflog` and `presvote`.

   I regress *voteshare* on *difflog* and *presvote* using the following commands:

```
reg_voteshare_difflog_presvote <- lm(inc.sub$voteshare ~ inc.sub$difflog
    + inc.sub$presvote, data = inc.sub)
reg_voteshare_difflog_presvote
summary(reg_voteshare_difflog_presvote)
```

   The "summary" command produces the following output:

```
Residuals:
Min       1Q   Median       3Q      Max
-0.25928 -0.04737 -0.00121  0.04618  0.33126

Coefficients:
Estimate Std. Error t value Pr(>|t|)
(Intercept)      0.4486442  0.0063297   70.88   <2e-16 ***
inc.sub$difflog  0.0355431  0.0009455   37.59   <2e-16 ***
inc.sub$presvote 0.2568770  0.0117637   21.84   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.07339 on 3190 degrees of freedom
Multiple R-squared:  0.4496, Adjusted R-squared:  0.4493
F-statistic:  1303 on 2 and 3190 DF,  p-value: < 2.2e-16
```

2. Write the prediction equation.

   From the output above:

   $\hat{\beta}_0 = 0.4486442$

   $\hat{\beta}_1 = 0.0355431$

   $\hat{\beta}_2 = 0.2568770$

   Therefore:

   $$\hat{voteshare} = \mathbf{0.4486442 + 0.0355431(\textit{difflog}) + 0.2568770(\textit{presvote})} + \epsilon$$

On average, holding presidential candidate vote share and the log of the difference in campaign spending constant at zero, we would expect the incumbent's vote share to be 0.4486442. On average, holding presidential candidate vote share constant at zero, we would expect a one-unit increase in the log of the difference in campaign spending to be associated with an increase of 0.0355431 in the incumbent's vote share. On average, holding the difference of the log in campaign spending constant at zero, we would expect a presidential candidate vote share increasing from 0 to 100 percent (i.e., increasing presidential candidate vote share from 0.0 to 1.0) to be associated with a 0.2568770 increase in the incumbent's vote share.

3. What is it in this output that is identical to the output in Question 4? Why do you think this is the case?

From the output tables above, I see that the minimum, first quartile, median, third quartile, and maximum residuals are identical:

```
Residuals:
Min       1Q   Median       3Q      Max
-0.25928 -0.04737 -0.00121  0.04618  0.33126
```

The residual standard errors are also identical:

```
Residual standard error: 0.07339
```

****** ACADEMIC INTEGRITY NOTE: I spent days wrestling with this last piece of the problem set, so I turned to Google for assistance. As with all previous assignments in this class and others, I did not use any generative AI, as I oppose AI in general.***

The regressions in problems 1 and 2 involved performing bivariate regressions and storing the residuals from those regressions in separate variables. The regression in problem 4 involved regressing those residuals on one another. Through this process, I partialled out the effects of the covariates on the response variable *(John R. Crooker, Ph. D., 2019, University of Nevada - Las Vegas)*.

By the Frisch-Waugh-Lovell Theorem *(StackOverflow)*, the residuals from the partialled-out residual-on-residual regression are identical to the residuals of the full, non-partialled multivariate regression performed in problem 5.

This theorem could be particularly powerful when it comes the underlying goals of quantitative social science – causal inference, drawing conclusions about a population from sample data – as the partialling-out approach provides a method for isolating the effect of a single covariate on the response variable.